

Student Name: _____

Class Name: **College Algebra FA26 - 014 MW 2:30**Number of Questions: **10**Instructor Name: **Mamani Velasco, Ruben****Question 1 of 10**

Consider the following quadratic function.

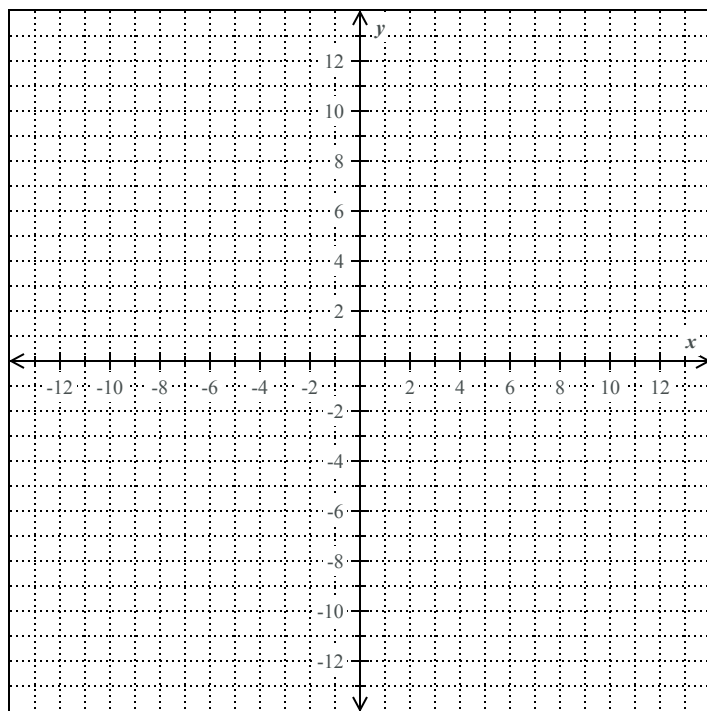
$$g(x) = -3x^2 + 12x - 11$$

- (a) Write the equation in the form $g(x) = a(x - h)^2 + k$. Then give the vertex of its graph.

Writing in the form specified: $g(x) =$

Vertex:

- (b) Graph the function.

**Question 2 of 10**

Rewrite the function $f(x) = -2(x + 1)^2 - 12$ in the form $f(x) = ax^2 + bx + c$.

Question 3 of 10

Find the x -intercept(s) and the coordinates of the vertex for the parabola $y = x^2 - 2x - 15$. If there is more than one x -intercept, separate them with commas.

x -intercept(s): _____

vertex: (_____ , _____)

Question 4 of 10

Answer the questions below about the quadratic function.

$$g(x) = 3x^2 + 18x + 25$$

(a) Does the function have a minimum or maximum value?

(b) What is the function's minimum or maximum value?

(c) Where does the minimum or maximum value occur?

$x =$ _____

Question 5 of 10

A ball is thrown vertically upward. After t seconds, its height h (in feet) is given by the function

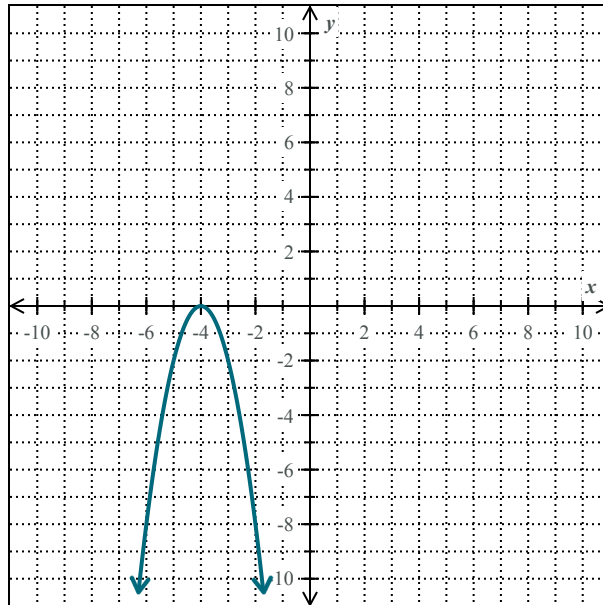
$h(t) = 112t - 16t^2$. After how long will it reach its maximum height?

Do not round your answer.

Time: _____ seconds

Question 6 of 10

The graph of a quadratic function with vertex $(-4, 0)$ is shown in the figure below. Find the domain and the range.



Write the domain and range using interval notation.

(a) domain = _____

(b) range = _____

Question 7 of 10

Find all the zeros of the quadratic function.

$$y = x^2 - 9x - 36$$

If there is more than one zero, separate them with commas.

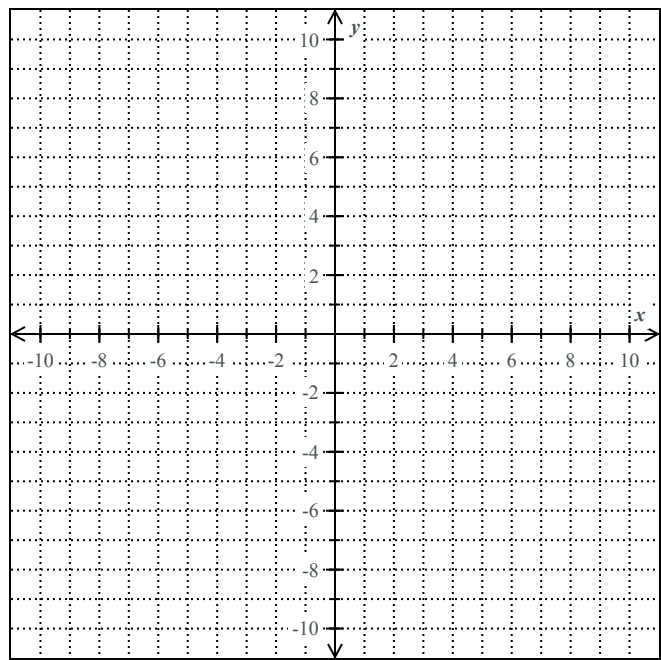
Question 8 of 10

Solve the equation by graphing.

$$-x^2 - 12x - 27 = 0$$

First, graph the associated parabola by plotting the vertex and four additional points, two on each side of the vertex.

Then, use the graph to give the solution(s) to the equation.
If there is more than one solution, separate them with commas.

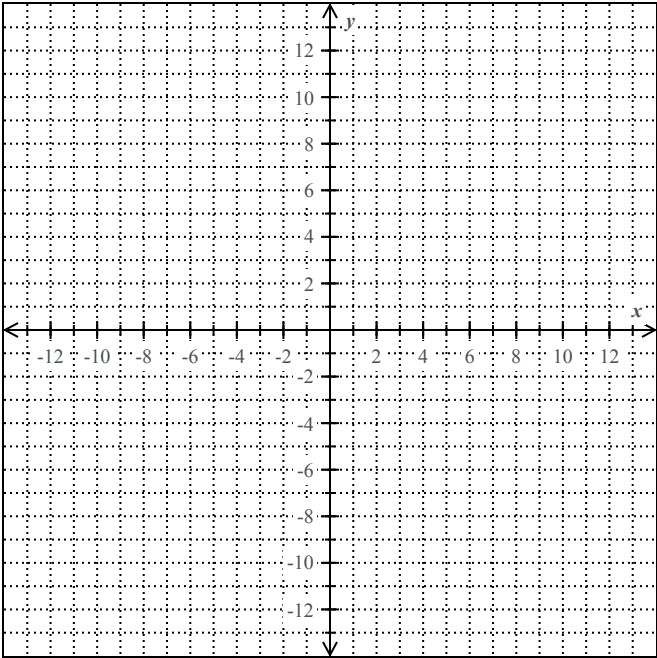


Solution(s):

Question 9 of 10

Graph the parabola.

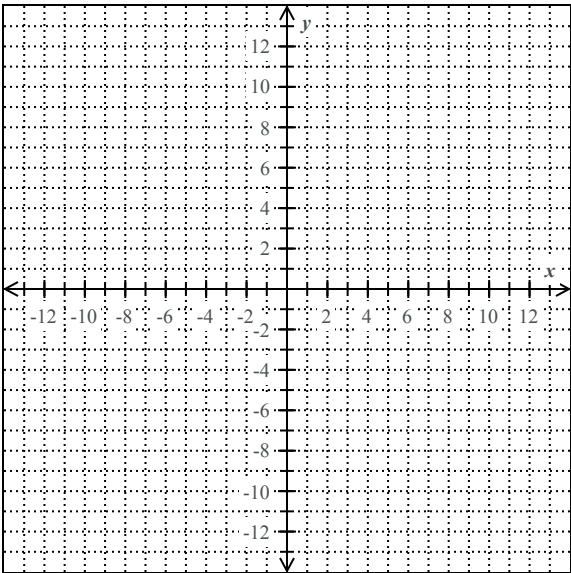
$$y = -3(x + 4)^2 - 1$$



Question 10 of 10

Graph the parabola.

$$y = -\frac{2}{3}x^2 - 4x - 4$$



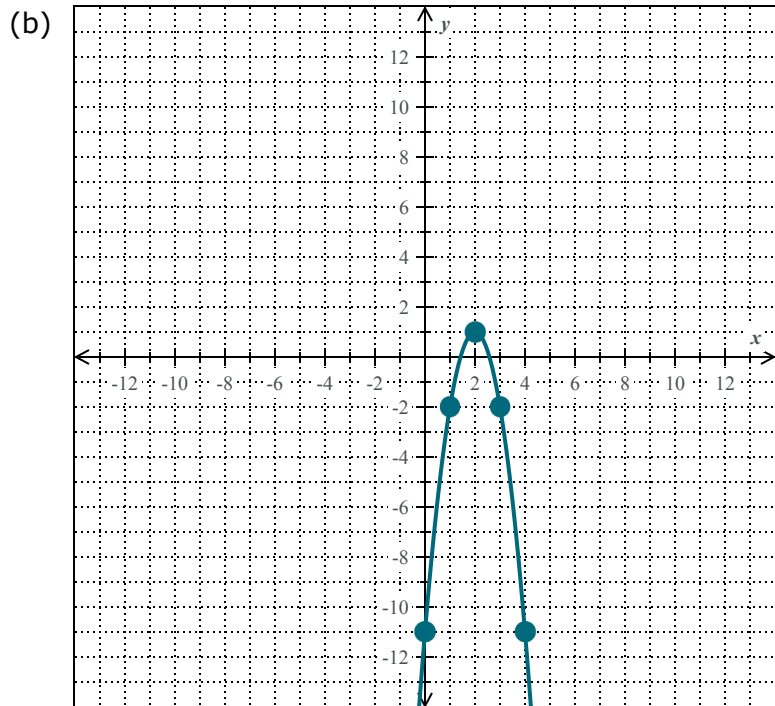
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Question 1 of 10

(a) Writing in the form specified: $g(x) = -3(x - 2)^2 + 1$

Vertex: $(2, 1)$



Question 2 of 10

$$f(x) = -2x^2 - 4x - 14$$

Question 3 of 10

x-intercept(s): $-3, 5$

vertex: $(1, -16)$

Question 4 of 10

(a) Does the function have a minimum or maximum value?

minimum

(b) What is the function's minimum or maximum value?

-2

(c) Where does the minimum or maximum value occur?

$x = -3$

Question 5 of 10

Time: 3.5 seconds

Question 6 of 10

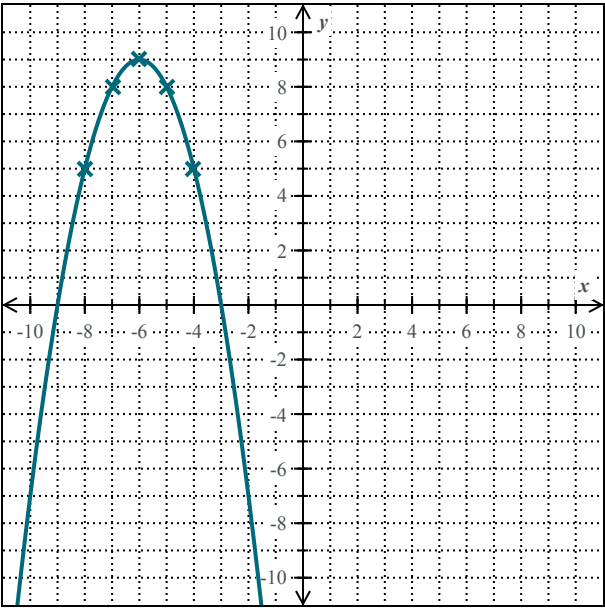
(a) domain: $(-\infty, \infty)$

(b) range: $(-\infty, 0]$

Question 7 of 10

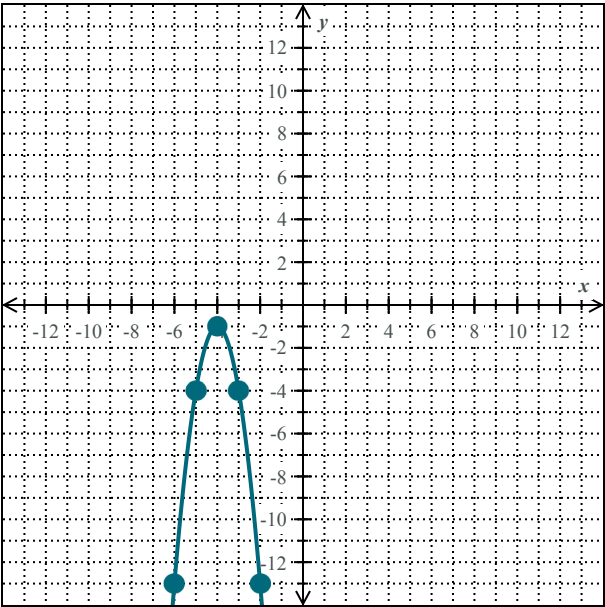
zero(s): $-3, 12$

Question 8 of 10



Solution(s): $x = -9, -3$

Question 9 of 10



Question 10 of 10

